Paper-II Set-A

1. Which of the following best describes encapsulation in OOP?

   A. Converting source code into machine code
   B. Bundling data and methods together and controlling access to them
   C. Creating multiple objects from different classes
   D. Executing multiple functions simultaneously

2. An object is best described as:

   A. A blueprint for creating classes
   B. An instance of a class
   C. A collection of functions only
   D. A compiler directive

3. Which OOP principle allows the same interface to represent different underlying implementations?

   A. Encapsulation
   B. Abstraction
   C. Polymorphism
   D. Aggregation

4. What is the primary purpose of a destructor?

   A. Allocate memory for an object
   B. Initialize data members
   C. Perform cleanup when an object is destroyed
   D. Overload operators

5. Which of the following is an example of function overloading?

   A. Redefining a class
   B. Defining a function only once
   C. Defining functions with the same name but different parameter lists
   D. Calling a derived-class function

6. A virtual function primarily supports:

   A. Compile-time memory allocation
   B. Static binding
   C. Runtime polymorphism
   D. Data hiding only

7. What is the major distinction between function overloading and function overriding?

   A. Both occur only at runtime
   B. Overloading uses different parameter lists, while overriding provides a derived-class implementation of an inherited virtual function
   C. Overloading requires inheritance, overriding does not
   D. Overriding is always compile-time polymorphism

8. If a base-class function is not declared virtual and a derived class defines a function with the same signature, calling through a base-class pointer generally results in:

A. Runtime dispatch to the derived function
B. Static binding to the base-class function
C. Compilation failure in all cases
D. Invocation of both functions

9. Which statement about a pure virtual function is correct?

A. It must always have a complete implementation in the base class
B. It cannot be inherited
C. It makes the class abstract when the class has at least one pure virtual function
D. It can only be declared private

10. Consider `Base *p = new Derived();`. If `Base` has a virtual destructor, deleting `p` will**:**

A. Invoke only the base destructor
B. Invoke only the derived destructor
C. Invoke the derived destructor followed by the base destructor
D. Always cause a compilation error

11. Which inheritance structure is most directly associated with the classic "diamond problem"?

A. Single inheritance
B. Hierarchical inheritance
C. Multiple inheritance through a common base
D. Multilevel inheritance without branching

12. Which of the following is a valid characteristic of operator overloading?

A. It can create completely new operators
B. Operator precedence can be changed
C. The overloaded operator retains its existing precedence and associativity
D. Every operator must be overloaded as a member function

13. In C++, if a class contains a pointer data member and its default copy constructor performs member-wise copying, the result can be:

A. Automatic garbage collection
B. Two objects pointing to the same dynamically allocated memory
C. Guaranteed deep copying
D. Automatic reference counting

14. Which operation is most appropriate for random access in a file?

A. Reading only from the beginning
B. Sequential character scanning only
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15. A virtual function call typically requires runtime support such as a:

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B. Preprocessor directive
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A. Operator precedence
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19. In runtime polymorphism, which combination is most appropriate for ensuring correct dynamic dispatch and cleanup through a base pointer?

A. Non-virtual function and non-virtual destructor
B. Static function and virtual destructor
C. Virtual overridden function and virtual destructor in the polymorphic base
D. Friend function and static destructor

20. Consider a class hierarchy where a base pointer refers to a derived object. If an overridden method is virtual, which principle determines the implementation executed?

A. Lexical scoping
B. Early binding
C. Dynamic binding
D. Operator precedence

WEB TECHNOLOGIES

21. DNS primarily translates:

A. MAC addresses into IP addresses
B. Domain names into IP addresses
C. HTTP requests into HTML
D. URLs into passwords

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A. `<section>`
B. `<nav>`
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D. CSSOM only

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32. RESTful APIs commonly identify resources using:

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B. URLs/URIs
C. HTML comments
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33. Which technology is specifically intended for persistent, bidirectional communication between a client and server?

A. FTP
B. SMTP
C. WebSocket
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34. Which Linux web server is widely used as a reverse proxy and HTTP server?

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37. The `SameSite` cookie attribute primarily helps mitigate:

A. SQL injection
B. Buffer overflow
C. Cross-site request forgery in applicable scenarios
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38. Cross-Site Scripting (XSS) occurs primarily when:

A. SQL queries are not indexed
B. Untrusted input is received as client-side code
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39. What is the key benefit of asynchronous JavaScript programming?

A. It eliminates HTTP and does not require server side scripting
B. It allows potentially long-running operations without unnecessarily blocking the main execution flow
C. It removes the need for servers
D. It converts JavaScript to SQL so that it can be executed from the client side

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A. It is always faster and more responsive
B. It cannot be bypassed merely by modifying or disabling browser-side code
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46. In an event-driven web server architecture, a major advantage of non-blocking I/O is that:

A. Every request requires a dedicated operating-system process
B. A relatively small number of execution resources can handle many I/O-bound operations concurrently
C. Database transactions become unnecessary
D. Network latency becomes zero

47. A web application stores an authentication token in a cookie configured with `HttpOnly`**,** `Secure`**,** and an appropriate `SameSite` policy**.** Which statement is most accurate?

A. XSS becomes impossible
B. CSRF becomes impossible in every deployment
C. These attributes reduce important attack surfaces but do not eliminate all web-application vulnerabilities
D. TLS is no longer necessary

48. A service receives the same payment request multiple times because of retries. Which API design property is most useful to prevent duplicate business effects?

A. Larger HTML pages
B. Random URLs
C. Idempotency using an appropriate idempotency key or equivalent mechanism
D. Disabling HTTP status codes

49. Which statement most accurately describes the role of a circuit breaker in microservices?

A. It encrypts database records
B. It limits repeated calls to an unhealthy dependency and allows controlled recovery attempts
C. It creates database indexes
D. It replaces load balancing

50. A frontend makes a cross-origin request using a non-simple HTTP method and custom headers. What is the browser likely to perform before the actual request?

A. DNS transfer
B. WebSocket upgrade
C. A CORS preflight request using OPTIONS
D. An FTP login

DATABASE MANAGEMENT SYSTEMS

51. A database schema represents:

A. Current data values only
B. A single query result
C. The logical structure/design of the database
D. A backup file only

52. A database instance refers to:

A. The database design at a particular point of time
B. The ER diagram only
C. The DBMS software source code
D. The data stored in the database at a particular point time

53. Which property allows changes in physical storage without requiring changes to the logical schema?

A. Logical data independence
B. Physical data independence
C. Referential integrity
D. Entity integrity

54. Which SQL clause filters groups after aggregation?

WHERE
B. ORDER BY
C. DISTINCT
D. HAVING

55. A foreign key is primarily used to enforce:

A. Encryption
B. Query optimization
C. Referential integrity
D. Data compression

56. Second Normal Form (2NF) in a database system removes:

A. Transitive dependencies only
B. Partial dependencies on a candidate key
C. All foreign keys
D. All functional dependencies

57. Third Normal Form (3NF) in a relational database primarily addresses:

A. Transitive dependency of non-key attributes on a key
B. Repeating groups
C. Partial dependency only
D. Physical storage fragmentation

58. Which SQL command permanently removes a table definition and its data?

A. DELETE
B. REMOVE
C. DROP
D. CLEAR

59. If $AB \rightarrow C$ and $A \rightarrow C$, then the dependency $AB \rightarrow C$ is:

A. Partially dependent on A with respect to composite determinant AB
B. A transitive dependency
C. A multivalued dependency
D. A join dependency

60. A relation is in 3NF if, for every non-trivial FD $X \rightarrow A$, at least one of the following holds:

A. X is always a foreign key
B. A is always a prime attribute
C. X is a superkey or A is a prime attribute
D. X contains exactly one attribute

61. Which statement about decomposition is correct?

A. Every decomposition is automatically lossless
B. A good decomposition should ideally be lossless and dependency-preserving where possible
C. Dependency preservation is always more important than losslessness
D. Normalization eliminates all joins

62. Which anomaly is most directly caused by storing the same fact redundantly in multiple rows?

A. DNS anomaly
B. Authentication anomaly
C. Update anomaly
D. Lock anomaly

63. Which relational algebra operation corresponds to selecting specific columns?

A. Selection
B. Projection
C. Join
D. Union

64. Which condition is required for a basic UNION operation between two relations?

A. They must have identical primary keys
B. They must have the same foreign keys
C. They must be union-compatible
D. They must have identical indexes

65. Which phenomenon occurs when a transaction reads data written by another transaction that has not yet committed?

A. Lost update
B. Dirty read
C. Phantom read
D. Cascading commit

66. Which isolation problem involves retrieving different sets of rows when the same range query is executed twice because another transaction inserted/deleted qualifying rows?

A. Dirty read
B. Lost update
C. Phantom read
D. Write skew only

67. Strict two-phase locking in a relational database provides an important benefit because:

A. It eliminates all deadlocks
B. It helps ensure conflict serializability and prevents certain cascading rollback problems by holding write locks until commit/abort
C. It eliminates the need for transactions
D. It guarantees no starvation

68. Which situation can produce a deadlock in DBMS?

A. Two transactions only reading data
B. T1 waits for a resource held by T2 while T2 waits for a resource held by T1
C. A transaction commits
D. A query uses an index

69. Which approach can be used for deadlock prevention?

Increasing table size
C. Removing all transactions
D. Disabling indexes
B. Imposing an ordering on resource acquisition

70. What is the primary purpose of an index in a relational database?

A. Guarantee normalization
B. Speed up certain data retrieval operations at additional storage/write-maintenance cost
C. Replace primary keys
D. Eliminate transactions

71. Which factor can make an index less useful for a query?

A. A predicate matching a very large proportion of the table
B. Highly selective predicate
C. Appropriate indexed column
D. Efficient query plan

72. In query optimization, a selection operation is often pushed closer to the base relation because it can:

Increase the number of intermediate tuples
B. Disable indexing
C. Reduce the amount of data processed by subsequent operations
D. Eliminate all joins

73. Which type of join returns all matching rows and additionally unmatched rows from the left relation?

A. Inner join
B. Left outer join
C. Right outer join
D. Cross join

74. A decomposition of R into R1 and R2 is lossless with respect to a set of Functional Dependencies when:

A. The natural join of the decomposed relations reconstructs exactly the original relation under the dependencies
B. Both relations have no common attributes
C. Every relation has a primary key
D. All dependencies are removed

75. A schedule has a cycle in its precedence graph. What can definitely be concluded?

A. It is not conflict-serializable
B. It is conflict-serializable
C. It is necessarily serial
D. It contains only read operations

76. Suppose T1 holds X-lock on item A and requests X-lock on B, while T2 holds X-lock on B and requests X-lock on A. The resulting condition is:

A. Cascading commit
B. Dirty read
C. Deadlock
D. Phantom read

77. Which schedule property ensures that committed database transactions do not depend on the results of transactions that subsequently abort?

A. Non-serializability
B. Recoverability
C. Denormalization
D. Fragmentation

78. Under strict two-phase locking, if a transaction has acquired an exclusive lock on X, another transaction generally cannot access X for conflicting operations until:

A. The first transaction performs a SELECT
B. Five operations have completed
C. The first transaction commits or aborts and releases the lock
D. The database is restarted

79. Which situation best illustrates a lost-update anomaly in DBMS?

A. Two transactions only read the same value
B. A transaction reads uncommitted data
C. A new row appears in a repeated range query
D. Two transactions read the same old value and later overwrite each other's updates

80. Which statement most accurately describes the relationship between serializability and recoverability in DBMS?

A. Serializability concerns correctness of concurrent execution order, while recoverability concerns safe commit/rollback dependencies
B. They are identical properties
C. Recoverability guarantees serial execution
D. Serializability guarantees durability after power failure

CLOUD COMPUTING

81. Which cloud service model provides virtualized computing resources such as virtual machines, storage and networks?

A. SaaS
B. IaaS
C. PaaS
D. DBaaS only

82. Which service model provides a platform on which developers can deploy applications without managing the underlying infrastructure directly?

A. IaaS
B. PaaS
C. SaaS
D. Bare metal

83. Which model provides complete software applications to end users over the network?

A. IaaS
B. PaaS
C. SaaS
D. HaaS

84. Serverless computing primarily means:

A. There are no servers anywhere
B. The cloud provider manages server infrastructure while users focus primarily on application code/services
C. Applications cannot scale due to absence of dedicated servers
D. Applications run only on distributed personal computers

85. Which virtualization type runs virtual machines directly on the physical hardware?

A. Type 2 hypervisor
B. Type 1 hypervisor
C. Container virtualization only
D. Application virtualization only

86. Docker in the cloud environment is primarily associated with:

A. Database normalization
B. Containerization
C. DNS management
D. HTML rendering

87. Which tool is commonly associated with infrastructure as code?

A. Photoshop
B. Terraform
C. HTML
D. SMTP

88. What is the primary purpose of a cloud load balancer?

A. Encrypt source code
B. Distribute incoming traffic across available resources
C. Balancing storage loads across devices
D. Compile containers

89. High availability of cloud infrastructure primarily means:

A. The system never needs maintenance
B. The system is designed to minimize service interruption
C. Every component is duplicated on the same disk to make data available anytime
D. The system uses only one server in order to reduce delay to communicate each other

90. Which cloud characteristic allows resources to be increased or decreased according to demand?

A. Immutability
B. Elasticity
C. Dynamics
D. Flexibility

91. IAM in Cloud Computing primarily deals with:

A. Image compression
B. Identity and access management
C. Application styling
D. CPU virtualization only

92. Kubernetes primarily uses which abstraction to provide a stable network endpoint for a changing set of Pods?

A. Volume
B. ConfigMap
C. Service
D. Namespace

93. Which mechanism is most appropriate for storing sensitive Kubernetes configuration such as credentials?

A. ConfigMap
B. Secret
C. Service
D. ReplicaSet only

94. Which cloud architecture is generally most resilient to failure of a single availability zone?

A. All components deployed in one zone
B. One server with no redundancy
C. Redundant components distributed across multiple availability zones
D. One container on one host

95. Which storage characteristic is generally associated with object storage?

A. It exposes only CPU registers
B. Data is stored as objects with associated metadata and identifiers
C. It requires every object to be a VM
D. It cannot scale horizontally

96. Which cloud security mechanism limits what an authenticated identity is permitted to do?

A. Encryption of personal access codes
B. Limitation on access to DNS
C. Authorization/IAM policy
D. Load balancing to prevent intrusion

97. Which practice best improves cloud disaster recovery readiness?

A. Keeping the only backup on the same failed system
B. Maintaining tested backups/recovery procedures and appropriately isolated recovery resources
C. Disabling monitoring to reduce the risk of exposure to unwanted personnel
D. Using one availability zone only

98. FinOps is primarily concerned with:

A. Database normalization and security achieved from it
B. Encrypting DNS so as to avoid intrusion
C. Managing and optimizing cloud costs in collaboration with experts
D. Container compilation

99. Which architecture is most suitable when an application needs to process events asynchronously at very high volume while decoupling producers and consumers?

A. Single monolithic process with blocking calls
B. Event-driven architecture using durable messaging infrastructure
C. Static HTML only
D. Direct database access from every client

100. A cloud application uses auto-scaling but experiences repeated scaling up and down during short workload fluctuations. Which design concern is most relevant?

A. Database normalization
B. Scaling hysteresis
C. HTML semantics
D. DNS naming only

Paper - II Set - B

1. What is the major distinction between function overloading and function overriding?

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48. A web application stores an authentication token in a cookie configured with `HttpOnly`, `Secure`, and an appropriate `SameSite` policy. Which statement is most accurate?

A. XSS becomes impossible
B. CSRF becomes impossible in every deployment
C. These attributes reduce important attack surfaces but do not eliminate all web-application vulnerabilities
D. TLS is no longer necessary

49. A service receives the same payment request multiple times because of retries. Which API design property is most useful to prevent duplicate business effects?

A. Larger HTML pages
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50. Which statement most accurately describes the role of a circuit breaker in microservices?

A. It encrypts database records
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WHERE
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C. SaaS
D. Bare metal

82. A cloud application uses auto-scaling but experiences repeated scaling up and down during short workload fluctuations. Which design concern is most relevant?

A. Database normalization
B. Scaling hysteresis
C. HTML semantics
D. DNS naming only

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A. SaaS
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C. PaaS
D. DBaaS only

92. Which practice best improves cloud disaster recovery readiness?

A. Keeping the only backup on the same failed system
B. Maintaining tested backups/recovery procedures and appropriately isolated recovery resources
C. Disabling monitoring to reduce the risk of exposure to unwanted personnel
D. Using one availability zone only

93. FinOps is primarily concerned with:

A. Database normalization and security achieved from it
B. Encrypting DNS so as to avoid intrusion
C. Managing and optimizing cloud costs in collaboration with experts
D. Container compilation

94. Which architecture is most suitable when an application needs to process events asynchronously at very high volume while decoupling producers and consumers?

A. Single monolithic process with blocking calls
B. Event-driven architecture using durable messaging infrastructure
C. Static HTML only
D. Direct database access from every client

95. IAM in Cloud Computing primarily deals with:

A. Image compression
B. Identity and access management
C. Application styling
D. CPU virtualization only

96. Kubernetes primarily uses which abstraction to provide a stable network endpoint for a changing set of Pods?

A. Volume
B. ConfigMap
C. Service
D. Namespace

97. Which mechanism is most appropriate for storing sensitive Kubernetes configuration such as credentials?

A. ConfigMap
B. Secret
C. Service
D. ReplicaSet only

98. Which cloud architecture is generally most resilient to failure of a single availability zone?

A. All components deployed in one zone
B. One server with no redundancy
C. Redundant components distributed across multiple availability zones
D. One container on one host

99. Which storage characteristic is generally associated with object storage?

A. It exposes only CPU registers
B. Data is stored as objects with associated metadata and identifiers
C. It requires every object to be a VM
D. It cannot scale horizontally

100. Which cloud security mechanism limits what an authenticated identity is permitted to do?

A. Encryption of personal access codes
B. Limitation on access to DNS
C. Authorization/IAM policy
D. Load balancing to prevent intrusion

Paper - II Set - C

1. Which OOP principle allows the same interface to represent different underlying implementations?

   A. Encapsulation
   B. Abstraction
   C. Polymorphism
   D. Aggregation

2. What is the primary purpose of a destructor?

   A. Allocate memory for an object
   B. Initialize data members
   C. Perform cleanup when an object is destroyed
   D. Overload operators

3. Which of the following is an example of function overloading?

   A. Redefining a class
   B. Defining a function only once
   C. Defining functions with the same name but different parameter lists
   D. Calling a derived-class function

4. A virtual function primarily supports:

   A. Compile-time memory allocation
   B. Static binding
   C. Runtime polymorphism
   D. Data hiding only

5. What is the major distinction between function overloading and function overriding?

   A. Both occur only at runtime
   B. Overloading uses different parameter lists, while overriding provides a derived-class implementation of an inherited virtual function
   C. Overloading requires inheritance, overriding does not
   D. Overriding is always compile-time polymorphism

6. If a base-class function is not declared virtual and a derived class defines a function with the same signature, calling through a base-class pointer generally results in:

   A. Runtime dispatch to the derived function
   B. Static binding to the base-class function
   C. Compilation failure in all cases
   D. Invocation of both functions

7. Which statement about a pure virtual function is correct?

   A. It must always have a complete implementation in the base class
   B. It cannot be inherited
   C. It makes the class abstract when the class has at least one pure virtual function
   D. It can only be declared private

8. Consider `Base *p = new Derived();`. If `Base` has a virtual destructor, deleting `p` will**:**

   A. Invoke only the base destructor
   B. Invoke only the derived destructor
   C. Invoke the derived destructor followed by the base destructor
   D. Always cause a compilation error

9. A virtual function call typically requires runtime support such as a:

   A. Macro table
   B. Preprocessor directive
   C. Virtual function table mechanism
   D. Header guard

10. Which statement about static data members of a class is correct?

    A. Each object necessarily has a separate copy
    B. They are shared among objects of the class
    C. They cannot be accessed by member functions
    D. They must always be dynamically allocated

11. Which inheritance structure is most directly associated with the classic "diamond problem"?

    A. Single inheritance
    B. Hierarchical inheritance
    C. Multiple inheritance through a common base
    D. Multilevel inheritance without branching

12. Which of the following is a valid characteristic of operator overloading?

    A. It can create completely new operators
    B. Operator precedence can be changed
    C. The overloaded operator retains its existing precedence and associativity
    D. Every operator must be overloaded as a member function

13. In C++, if a class contains a pointer data member and its default copy constructor performs member-wise copying, the result can be:

    A. Automatic garbage collection
    B. Two objects pointing to the same dynamically allocated memory
    C. Guaranteed deep copying
    D. Automatic reference counting

14. Which of the following best describes encapsulation in OOP?
A. Converting source code into machine code
B. Bundling data and methods together and controlling access to them
C. Creating multiple objects from different classes
D. Executing multiple functions simultaneously

15. An object is best described as:
A. A blueprint for creating classes
B. An instance of a class
C. A collection of functions only
D. A compiler directive

16. Which operation is most appropriate for random access in a file?

A. Reading only from the beginning
B. Sequential character scanning only
C. Moving the file position indicator to a required location
D. Reopening the file after every byte

17. In a hierarchy using virtual inheritance, the virtual base class sub-object is initialized by:

A. Every intermediate class independently
B. The first class appearing in the inheritance list
C. The most-derived class
D. The compiler-generated destructor

18. Suppose a class has a dynamically allocated resource and defines a destructor but does not correctly define/copy or move the resource ownership semantics. Which issue is most closely associated with the Rule of Three/Rule of Five?

A. Operator precedence
B. Static binding
C. Incorrect resource ownership during copying/moving
D. Multiple inheritance ambiguity only

19. Consider a class hierarchy where a base pointer refers to a derived object. If an overridden method is virtual, which principle determines the implementation executed?

A. Lexical scoping
B. Early binding
C. Dynamic binding
D. Operator precedence

20. In runtime polymorphism, which combination is most appropriate for ensuring correct dynamic dispatch and cleanup through a base pointer?

A. Non-virtual function and non-virtual destructor
B. Static function and virtual destructor
C. Virtual overridden function and virtual destructor in the polymorphic base
D. Friend function and static destructor

21. Responsive web design aims to:

A. Respond to the request made by the user and return results
B. Adapt the presentation to different screen sizes and devices
C. Disable JavaScript
D. Replace HTTP with FTP

22. Which JavaScript API provides access to the HTML document structure?

A. BOM only
B. DOM
C. SQL DOM
D. CSSOM only

23. JSON is primarily used in Web Technology as a:

A. Programming language
B. Data interchange format
C. Web server
D. Database engine

24. Which HTTP method is conventionally used to retrieve a resource?

A. GET
B. POST
C. PATCH
D. DELETE

25. Which HTTP status code generally indicates "Not Found"?

A. 200
B. 301
C. 404
D. 403

26. In a typical frontend framework, routing is used primarily to:

A. Encrypt passwords
B. Map URLs or application states to views/components
C. Store relational data
D. Configure DNS

27. Which HTML element represents the primary navigation section of a page?

A. `<section>`
B. `<nav>`
C. `<aside>`
D. `<footer>`

28. DNS primarily translates:
A. MAC addresses into IP addresses
B. Domain names into IP addresses
C. HTTP requests into HTML
D. URLs into passwords

29. Which CSS layout model is particularly suited for one-dimensional layouts?
A. Float
B. Flexbox
C. Table layout only
D. Positioning only

30. Middleware in a web application commonly:

A. Replaces the database
B. Processes requests/responses between different stages of an application
C. Creates HTML tags
D. Compiles CSS

31. Which HTTP method is generally intended to create a new resource?

A. GET
B. POST
C. DELETE
D. HEAD

32. RESTful APIs commonly identify resources using:

A. CSS classes
B. URLs/URIs
C. HTML comments
D. DNS TXT records only

33. Which technology is specifically intended for persistent, bidirectional communication between a client and server?

A. FTP
B. SMTP
C. WebSocket
D. DNS

34. Which Linux web server is widely used as a reverse proxy and HTTP server?
A. MySQL
B. Docker
C. Nginx
D. Redis

35. Which HTTP method is generally idempotent but not safe?

A. GET
B. PUT
C. POST
D. CONNECT

36. Which HTTP status code most appropriately indicates that a request succeeded and created a new resource?

A. 200
B. 201
C. 204
D. 304

37. The `SameSite` cookie attribute primarily helps mitigate:

A. SQL injection
B. Buffer overflow
C. Cross-site request forgery in applicable scenarios
D. DNS poisoning

38. A web application stores an authentication token in a cookie configured with `HttpOnly`**,** `Secure`**,** and an appropriate `SameSite` policy**.** Which statement is most accurate?

A. XSS becomes impossible
B. CSRF becomes impossible in every deployment
C. These attributes reduce important attack surfaces but do not eliminate all web-application vulnerabilities
D. TLS is no longer necessary

39. Which Git operation creates a new branch pointer from the current commit?

A. git delete
B. git branch
C. git compile
D. git archive-only

40. A frontend makes a cross-origin request using a non-simple HTTP method and custom headers. What is the browser likely to perform before the actual request?

A. DNS transfer
B. WebSocket upgrade
C. A CORS preflight request using OPTIONS
D. An FTP login

41. Which backend framework/runtime combination is specifically JavaScript-based?

A. Django/Python
B. Laravel/PHP
C. Node.js/Express
D. Spring/Java only

42. Which HTTP header is most directly used to indicate the media type of the request or response body?

A. Host
B. Content-Type
C. Referer
D. Connection

43. What is the purpose of an API authentication token?

A. Define CSS inheritance
B. Allocate RAM and processor slot
C. Provide credentials or a security assertion that can be validated by the API
D. Resolve a hostname so as to point exactly the intended host

44. Which HTTP response is most appropriate when the server successfully processes a request but intentionally returns no response body?

A. 201
B. 204
C. 301
D. 404

45. A major advantage of server-side validation over client-side-only validation is that:

A. It is always faster and more responsive
B. It cannot be bypassed merely by modifying or disabling browser-side code
C. It eliminates the need for databases
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48. Cross-Site Scripting (XSS) occurs primarily when:

A. SQL queries are not indexed
B. Untrusted input is received as client-side code
C. DNS records expire
D. Cookies are encrypted

49. What is the key benefit of asynchronous JavaScript programming?

A. It eliminates HTTP and does not require server side scripting
B. It allows potentially long-running operations without unnecessarily blocking the main execution flow
C. It removes the need for servers
D. It converts JavaScript to SQL so that it can be executed from the client side

50. In an event-driven web server architecture, a major advantage of non-blocking I/O is that:

A. Every request requires a dedicated operating-system process
B. A relatively small number of execution resources can handle many I/O-bound operations concurrently
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B. Encrypting DNS so as to avoid intrusion
C. Managing and optimizing cloud costs in collaboration with experts
D. Container compilation

95. Which architecture is most suitable when an application needs to process events asynchronously at very high volume while decoupling producers and consumers?

A. Single monolithic process with blocking calls
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C. Static HTML only
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98. Which cloud architecture is generally most resilient to failure of a single availability zone?

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99. Which storage characteristic is generally associated with object storage?

A. It exposes only CPU registers
B. Data is stored as objects with associated metadata and identifiers
C. It requires every object to be a VM
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100. Which practice best improves cloud disaster recovery readiness?

A. Keeping the only backup on the same failed system
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D. Using one availability zone only

Paper - II Set - D

1. Which of the following is an example of function overloading?

   A. Redefining a class
   B. Defining a function only once
   C. Defining functions with the same name but different parameter lists
   D. Calling a derived-class function

2. Which OOP principle allows the same interface to represent different underlying implementations?

   A. Encapsulation
   B. Abstraction
   C. Polymorphism
   D. Aggregation

3. What is the primary purpose of a destructor?

   A. Allocate memory for an object
   B. Initialize data members
   C. Perform cleanup when an object is destroyed
   D. Overload operators

4. A virtual function primarily supports:

   A. Compile-time memory allocation
   B. Static binding
   C. Runtime polymorphism
   D. Data hiding only

5. What is the major distinction between function overloading and function overriding?

   A. Both occur only at runtime
   B. Overloading uses different parameter lists, while overriding provides a derived-class implementation of an inherited virtual function
   C. Overloading requires inheritance, overriding does not
   D. Overriding is always compile-time polymorphism

6. Which inheritance structure is most directly associated with the classic "diamond problem"?

   A. Single inheritance
   B. Hierarchical inheritance
   C. Multiple inheritance through a common base
   D. Multilevel inheritance without branching

7. Which of the following is a valid characteristic of operator overloading?

   A. It can create completely new operators
   B. Operator precedence can be changed
   C. The overloaded operator retains its existing precedence and associativity
   D. Every operator must be overloaded as a member function

8. In C++, if a class contains a pointer data member and its default copy constructor performs member-wise copying, the result can be:

   A. Automatic garbage collection
   B. Two objects pointing to the same dynamically allocated memory
   C. Guaranteed deep copying
   D. Automatic reference counting

9. Which of the following best describes encapsulation in OOP?

   A. Converting source code into machine code
   B. Bundling data and methods together and controlling access to them
   C. Creating multiple objects from different classes
   D. Executing multiple functions simultaneously

10. An object is best described as:

    A. A blueprint for creating classes
    B. An instance of a class
    C. A collection of functions only
    D. A compiler directive

11. Which operation is most appropriate for random access in a file?

    A. Reading only from the beginning
    B. Sequential character scanning only
    C. Moving the file position indicator to a required location
    D. Reopening the file after every byte

12. In a hierarchy using virtual inheritance, the virtual base class sub-object is initialized by:

    A. Every intermediate class independently
    B. The first class appearing in the inheritance list
    C. The most-derived class
    D. The compiler-generated destructor

13. Suppose a class has a dynamically allocated resource and defines a destructor but does not correctly define/copy or move the resource ownership semantics. Which issue is most closely associated with the Rule of Three/Rule of Five?

    A. Operator precedence
    B. Static binding
    C. Incorrect resource ownership during copying/moving
    D. Multiple inheritance ambiguity only

14. Consider a class hierarchy where a base pointer refers to a derived object. If an overridden method is virtual, which principle determines the implementation executed?
A. Lexical scoping
B. Early binding
C. Dynamic binding
D. Operator precedence

15. If a base-class function is not declared virtual and a derived class defines a function with the same signature, calling through a base-class pointer generally results in:
A. Runtime dispatch to the derived function
B. Static binding to the base-class function
C. Compilation failure in all cases
D. Invocation of both functions

16. JSON is primarily used in Web Technology as a:

A. Programming language
B. Data interchange format
C. Web server
D. Database engine

17. Which HTTP method is conventionally used to retrieve a resource?

A. GET
B. POST
C. PATCH
D. DELETE

18. Which statement about a pure virtual function is correct?

A. It must always have a complete implementation in the base class
B. It cannot be inherited
C. It makes the class abstract when the class has at least one pure virtual function
D. It can only be declared private

19. Consider `Base *p = new Derived();`. If `Base` has a virtual destructor, deleting `p` will**:**

A. Invoke only the base destructor
B. Invoke only the derived destructor
C. Invoke the derived destructor followed by the base destructor
D. Always cause a compilation error

20. A virtual function call typically requires runtime support such as a:

A. Macro table
B. Preprocessor directive
C. Virtual function table mechanism
D. Header guard

21. Which statement about static data members of a class is correct?

A. Each object necessarily has a separate copy
B. They are shared among objects of the class
C. They cannot be accessed by member functions
D. They must always be dynamically allocated

22. In runtime polymorphism, which combination is most appropriate for ensuring correct dynamic dispatch and cleanup through a base pointer?

A. Non-virtual function and non-virtual destructor
B. Static function and virtual destructor
C. Virtual overridden function and virtual destructor in the polymorphic base
D. Friend function and static destructor

23. Responsive web design aims to:

A. Respond to the request made by the user and return results
B. Adapt the presentation to different screen sizes and devices
C. Disable JavaScript
D. Replace HTTP with FTP

24. Which JavaScript API provides access to the HTML document structure?

A. BOM only
B. DOM
C. SQL DOM
D. CSSOM only

25. Which HTTP status code generally indicates "Not Found"?

A. 200
B. 301
C. 404
D. 403

26. In a typical frontend framework, routing is used primarily to:

A. Encrypt passwords
B. Map URLs or application states to views/components
C. Store relational data
D. Configure DNS

27. Which HTML element represents the primary navigation section of a page?

A. `<section>`
B. `<nav>`
C. `<aside>`
D. `<footer>`

28. DNS primarily translates:

A. MAC addresses into IP addresses
B. Domain names into IP addresses
C. HTTP requests into HTML
D. URLs into passwords

29. Middleware in a web application commonly:

A. Replaces the database
B. Processes requests/responses between different stages of an application
C. Creates HTML tags
D. Compiles CSS

30. Which HTTP method is generally intended to create a new resource?

A. GET
B. POST
C. DELETE
D. HEAD

31. RESTful APIs commonly identify resources using:

A. CSS classes
B. URLs/URIs
C. HTML comments
D. DNS TXT records only

32. Which technology is specifically intended for persistent, bidirectional communication between a client and server?

A. FTP
B. SMTP
C. WebSocket
D. DNS

33. Which CSS layout model is particularly suited for one-dimensional layouts?

A. Float
B. Flexbox
C. Table layout only
D. Positioning only

34. Which HTTP status code most appropriately indicates that a request succeeded and created a new resource?

A. 200
B. 201
C. 204
D. 304

35. The `SameSite` cookie attribute primarily helps mitigate:

A. SQL injection
B. Buffer overflow
C. Cross-site request forgery in applicable scenarios
D. DNS poisoning

36. A web application stores an authentication token in a cookie configured with `HttpOnly`, `Secure`, and an appropriate `SameSite` policy. Which statement is most accurate?

A. XSS becomes impossible
B. CSRF becomes impossible in every deployment
C. These attributes reduce important attack surfaces but do not eliminate all web-application vulnerabilities
D. TLS is no longer necessary

37. Which Git operation creates a new branch pointer from the current commit?

A. git delete
B. git branch
C. git compile
D. git archive-only

38. A frontend makes a cross-origin request using a non-simple HTTP method and custom headers. What is the browser likely to perform before the actual request?
A. DNS transfer
B. WebSocket upgrade
C. A CORS preflight request using OPTIONS
D. An FTP login

39. Which backend framework/runtime combination is specifically JavaScript-based?
A. Django/Python
B. Laravel/PHP
C. Node.js/Express
D. Spring/Java only

40. Which HTTP header is most directly used to indicate the media type of the request or response body?

A. Host
B. Content-Type
C. Referer
D. Connection

41. What is the purpose of an API authentication token?

A. Define CSS inheritance
B. Allocate RAM and processor slot
C. Provide credentials or a security assertion that can be validated by the API
D. Resolve a hostname so as to point exactly the intended host

42. Which Linux web server is widely used as a reverse proxy and HTTP server?
A. MySQL
B. Docker
C. Nginx
D. Redis

43. Which HTTP method is generally idempotent but not safe?
A. GET
B. PUT
C. POST
D. CONNECT

44. Which statement most accurately describes the role of a circuit breaker in microservices?

A. It encrypts database records
B. It limits repeated calls to an unhealthy dependency and allows controlled recovery attempts
C. It creates database indexes
D. It replaces load balancing

45. Cross-Site Scripting (XSS) occurs primarily when:

A. SQL queries are not indexed
B. Untrusted input is received as client-side code
C. DNS records expire
D. Cookies are encrypted

46. What is the key benefit of asynchronous JavaScript programming?

A. It eliminates HTTP and does not require server side scripting
B. It allows potentially long-running operations without unnecessarily blocking the main execution flow
C. It removes the need for servers
D. It converts JavaScript to SQL so that it can be executed from the client side

47. In an event-driven web server architecture, a major advantage of non-blocking I/O is that:

A. Every request requires a dedicated operating-system process
B. A relatively small number of execution resources can handle many I/O-bound operations concurrently
C. Database transactions become unnecessary
D. Network latency becomes zero

48. Which HTTP response is most appropriate when the server successfully processes a request but intentionally returns no response body?

A. 201
B. 204
C. 301
D. 404

49. A major advantage of server-side validation over client-side-only validation is that:

A. It is always faster and more responsive
B. It cannot be bypassed merely by modifying or disabling browser-side code
C. It eliminates the need for databases
D. It requires no authentication

50. A service receives the same payment request multiple times because of retries. Which API design property is most useful to prevent duplicate business effects?

A. Larger HTML pages
B. Random URLs
C. Idempotency using an appropriate idempotency key or equivalent mechanism
D. Disabling HTTP status codes

51. Second Normal Form (2NF) in a database system removes:

A. Transitive dependencies only
B. Partial dependencies on a candidate key
C. All foreign keys
D. All functional dependencies

52. A foreign key is primarily used to enforce:

A. Encryption
B. Query optimization
C. Referential integrity
D. Data compression

53. Third Normal Form (3NF) in a relational database primarily addresses:

A. Transitive dependency of non-key attributes on a key
B. Repeating groups
C. Partial dependency only
D. Physical storage fragmentation

54. A relation is in 3NF if, for every non-trivial FD $X \rightarrow A$, at least one of the following holds:

A. X is always a foreign key
B. A is always a prime attribute
C. X is a superkey or A is a prime attribute
D. X contains exactly one attribute

55. Which statement about decomposition is correct?

A. Every decomposition is automatically lossless
B. A good decomposition should ideally be lossless and dependency-preserving where possible
C. Dependency preservation is always more important than losslessness
D. Normalization eliminates all joins

56. Which condition is required for a basic UNION operation between two relations?

A. They must have identical primary keys
B. They must have the same foreign keys
C. They must be union-compatible
D. They must have identical indexes

57. Which phenomenon occurs when a transaction reads data written by another transaction that has not yet committed?

A. Lost update
B. Dirty read
C. Phantom read
D. Cascading commit

58. Which situation can produce a deadlock in DBMS?

A. Two transactions only reading data
B. T1 waits for a resource held by T2 while T2 waits for a resource held by T1
C. A transaction commits
D. A query uses an index

59. Which SQL command permanently removes a table definition and its data?

A. DELETE
B. REMOVE
C. DROP
D. CLEAR

60. If $AB \rightarrow C$ and $A \rightarrow C$, then the dependency $AB \rightarrow C$ is:

A. Partially dependent on A with respect to composite determinant AB
B. A transitive dependency
C. A multivalued dependency
D. A join dependency

61. Which anomaly is most directly caused by storing the same fact redundantly in multiple rows?

A. DNS anomaly
B. Authentication anomaly
C. Update anomaly
D. Lock anomaly

62. Which relational algebra operation corresponds to selecting specific columns?

A. Selection
B. Projection
C. Join
D. Union

63. A database schema represents:

A. Current data values only
B. A single query result
C. The logical structure/design of the database
D. A backup file only

64. Which SQL clause filters groups after aggregation?

WHERE
B. ORDER BY
C. DISTINCT
D. HAVING

65. Which isolation problem involves retrieving different sets of rows when the same range query is executed twice because another transaction inserted/deleted qualifying rows?

A. Dirty read
B. Lost update
C. Phantom read
D. Write skew only

66. Strict two-phase locking in a relational database provides an important benefit because:

A. It eliminates all deadlocks
B. It helps ensure conflict serializability and prevents certain cascading rollback problems by holding write locks until commit/abort
C. It eliminates the need for transactions
D. It guarantees no starvation

67. Which approach can be used for deadlock prevention?

Increasing table size
C. Removing all transactions
D. Disabling indexes
B. Imposing an ordering on resource acquisition

68. A database instance refers to:

A. The database design at a particular point of time
B. The ER diagram only
C. The DBMS software source code
D. The data stored in the database at a particular point time

69. Which situation best illustrates a lost-update anomaly in DBMS?

A. Two transactions only read the same value
B. A transaction reads uncommitted data
C. A new row appears in a repeated range query
D. Two transactions read the same old value and later overwrite each other's updates

70. What is the primary purpose of an index in a relational database?

A. Guarantee normalization
B. Speed up certain data retrieval operations at additional storage/write-maintenance cost
C. Replace primary keys
D. Eliminate transactions

71. Which factor can make an index less useful for a query?

A. A predicate matching a very large proportion of the table
B. Highly selective predicate
C. Appropriate indexed column
D. Efficient query plan

72. In query optimization, a selection operation is often pushed closer to the base relation because it can:

Increase the number of intermediate tuples
B. Disable indexing
C. Reduce the amount of data processed by subsequent operations
D. Eliminate all joins

73. Which type of join returns all matching rows and additionally unmatched rows from the left relation?

A. Inner join
B. Left outer join
C. Right outer join
D. Cross join

74. Which statement most accurately describes the relationship between serializability and recoverability in DBMS?

A. Serializability concerns correctness of concurrent execution order, while recoverability concerns safe commit/rollback dependencies
B. They are identical properties
C. Recoverability guarantees serial execution
D. Serializability guarantees durability after power failure

75. A decomposition of R into R1 and R2 is lossless with respect to a set of Functional Dependencies when:

A. The natural join of the decomposed relations reconstructs exactly the original relation under the dependencies
B. Both relations have no common attributes
C. Every relation has a primary key
D. All dependencies are removed

76. A schedule has a cycle in its precedence graph. What can definitely be concluded?

A. It is not conflict-serializable
B. It is conflict-serializable
C. It is necessarily serial
D. It contains only read operations

77. Suppose T1 holds X-lock on item A and requests X-lock on B, while T2 holds X-lock on B and requests X-lock on A. The resulting condition is:

A. Cascading commit
B. Dirty read
C. Deadlock
D. Phantom read

78. Which schedule property ensures that committed database transactions do not depend on the results of transactions that subsequently abort?

A. Non-serializability
B. Recoverability
C. Denormalization
D. Fragmentation

79. Which property allows changes in physical storage without requiring changes to the logical schema?

A. Logical data independence
B. Physical data independence
C. Referential integrity
D. Entity integrity

80. Under strict two-phase locking, if a transaction has acquired an exclusive lock on X, another transaction generally cannot access X for conflicting operations until:

A. The first transaction performs a SELECT
B. Five operations have completed
C. The first transaction commits or aborts and releases the lock
D. The database is restarted

81. Which cloud security mechanism limits what an authenticated identity is permitted to do?

A. Encryption of personal access codes
B. Limitation on access to DNS
C. Authorization/IAM policy
D. Load balancing to prevent intrusion

82. Which service model provides a platform on which developers can deploy applications without managing the underlying infrastructure directly?

A. IaaS
B. PaaS
C. SaaS
D. Bare metal

83. Which model provides complete software applications to end users over the network?

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B. PaaS
C. SaaS
D. HaaS

84. IAM in Cloud Computing primarily deals with:

A. Image compression
B. Identity and access management
C. Application styling
D. CPU virtualization only

85. Kubernetes primarily uses which abstraction to provide a stable network endpoint for a changing set of Pods?

A. Volume
B. ConfigMap
C. Service
D. Namespace

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88. Which cloud service model provides virtualized computing resources such as virtual machines, storage and networks?

A. SaaS
B. IaaS
C. PaaS
D. DBaaS only

89. Serverless computing primarily means:

A. There are no servers anywhere
B. The cloud provider manages server infrastructure while users focus primarily on application code/services
C. Applications cannot scale due to absence of dedicated servers
D. Applications run only on distributed personal computers

90. Docker in the cloud environment is primarily associated with:

A. Database normalization
B. Containerization
C. DNS management
D. HTML rendering

91. Which tool is commonly associated with infrastructure as code?

A. Photoshop
B. Terraform
C. HTML
D. SMTP

92. What is the primary purpose of a cloud load balancer?

A. Encrypt source code
B. Distribute incoming traffic across available resources
C. Balancing storage loads across devices
D. Compile containers

93. High availability of cloud infrastructure primarily means:

A. The system never needs maintenance
B. The system is designed to minimize service interruption
C. Every component is duplicated on the same disk to make data available anytime
D. The system uses only one server in order to reduce delay to communicate each other

94. Which virtualization type runs virtual machines directly on the physical hardware?

A. Type 2 hypervisor
B. Type 1 hypervisor
C. Container virtualization only
D. Application virtualization only

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